

# Sale Season

## Open questions and feature extension ideas

Status as of 4 September 2026 | commit a1b4e02 | live at sales.pfeifhofer.dev (NAS, port 8086)

### Where the project stands

All three planned milestones are built: the calendar, email reminders, and the installable app with push notifications. 51 tests pass and the production build is clean.

Two things have never been exercised for real. The Docker build has never run, because there is no Docker on the development machine - the first true test is the deploy itself. And no push notification has yet travelled through a real push service to a real phone.

The single most time-sensitive item in this document is Q1: published sale dates only cover 2026.

## 1. Open questions

These are decisions rather than tasks. Each one changes what gets built, so none of them should be answered by guessing on your behalf.

### Q1 How should the yearly date refresh happen?

Only six events carry published, verified dates - the four South Tyrol windows and the two national Italian saldi - and all six are for 2026. From January 2027 every regulated event silently falls back to its estimated recurrence rule. The backend now warns at startup and reports `datesNeedRefresh` on `/api/health`, so it cannot rot unnoticed, but nothing fixes it automatically. Italian regions publish winter dates in late December and summer dates in June; South Tyrol publishes to the Bolzano Chamber of Commerce page.

|                  |                                                                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Manual</b>    | You check the sources each December and June and I add the overrides. Simplest, most accurate, needs you to remember.                            |
| <b>Reminder</b>  | A scheduled task nags you twice a year when the dates are due. Cheap to add.                                                                     |
| <b>Automated</b> | Scrape the Chamber of Commerce and regional pages. Fragile, and a wrong scrape silently corrupts the one thing this project exists to get right. |

### Q2 Do the deal sites stay as plain links?

Geizhals, idealo, Toppreise, Trovaprezzi, Preisjaeger, mydealz and Preispirat are currently plain outbound links with no affiliate or referral parameters, and the interface says so explicitly. Going further is a commercial decision, not a technical one.

|                   |                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Keep as is</b> | Zero obligations, nothing to disclose, no maintenance.                                                                                               |
| <b>Live deals</b> | mydealz and Preisjaeger are Pepper-network sites with per-group RSS. A "hot deals right now" strip is genuinely cheap to build from that.            |
| <b>Affiliate</b>  | idealo and Geizhals run affiliate programmes with product feeds. Revenue, but it requires disclosure and changes the site from neutral to monetised. |

### Q3 Is this a private tool or a public site?

It is already publicly reachable and it collects email addresses. If it stays effectively personal, nothing changes. If you intend other people to use it, an Austrian/EU site collecting email addresses normally wants an Impressum and a privacy policy, and the subscription data becomes personal data you are responsible for.

|                 |                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------|
| <b>Personal</b> | Leave as is. The double opt-in and one-click unsubscribe already exceed what a personal tool needs. |
| <b>Public</b>   | Add Impressum and privacy pages, and decide on a retention policy for unconfirmed subscriptions.    |

### Q4 Should reminders be per-sale or a digest?

Right now each sale generates its own email at each configured lead time. With the default of 7 and 1 days that is roughly two emails per sale, and November is busy - Black Week, Black Friday, Cyber Monday and Singles Day cluster within three weeks and would produce a burst.

|                 |                                                                                                                             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Per-sale</b> | Current behaviour. Timely, but bursty in peak season. Multiple sales due on the same day are already merged into one email. |
| <b>Digest</b>   | One weekly email listing everything coming up. Calmer, but loses the "starts tomorrow" urgency.                             |
| <b>Both</b>     | Offer it as a choice at signup. More code, and another preference to explain.                                               |

### Q5 Which countries come next, if any?

The data model handles regulated, traditional and retailer-set dates, which is exactly what a wider European rollout needs. France and Belgium are strongly regulated with genuinely fixed national dates, so they would slot in cleanly and would show off the precision model better than the current four do.

|                     |                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| <b>Stay at four</b> | AT, DE, CH, IT covers where you actually shop.                                                      |
| <b>Add FR/BE</b>    | Both publish binding national dates. High data quality for modest effort.                           |
| <b>Wider</b>        | NL, ES, PL and others are mostly deregulated, so they would add traditional-precision entries only. |

## 2. Known gaps and risks

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Recorded in full in docs/KNOWN\_ISSUES.md. The items below are the ones with real consequences.

### No backup of the subscriber data

backend/data holds the subscriber list, the push device registry and the delivery log, as plain JSON on a single NAS volume with no backup and no export. Losing that volume loses every subscription with no way to rebuild it, and no way to tell anyone why their reminders stopped. This is the most serious operational gap in the project and is not yet addressed anywhere.

### Docker build has never been executed

Neither Python nor Docker is installed on the development machine. Both Dockerfiles and the compose wiring are therefore unverified by execution. The likeliest failure point is the multi-stage step in backend/Dockerfile that generates the dataset snapshot before the runtime stage copies it in.

### Push has never reached a real device

Endpoints, storage, expiry pruning and the reminder job's push branch were all exercised locally against fabricated subscriptions, and a dry run correctly pushed the South Tyrol winter sale to an Italian device in Italian while skipping the Austrian one. But nothing has travelled through Google's push service to an actual phone. That needs the deployed HTTPS site plus a handset.

### Light mode has had no human review

The light palette was built by inverting the ink scale behind CSS variables and verified programmatically for contrast - body text 17.0:1, buttons 9.0:1, muted text 4.8:1, all above the accessibility threshold. But nobody has looked at it. The year-timeline bars are the most likely problem: they are drawn from category colours at low alpha and were tuned against a dark background.

### Italian data is flattened in two places

The national saldi entries store one end date - the latest of any region - because ends vary from late February to early March. And Trentino, which sets dates by district, is not modelled at all and currently falls under the national entry, which is wrong in detail. South Tyrol is modelled correctly and separately.

### The preferences link is a capability

The preferences page is keyed by the unsubscribe token, so anyone holding a forwarded reminder email can change lead times and categories, not just unsubscribe. That was a deliberate trade to avoid inventing a second secret or a password, but it is worth knowing.

## 3. Feature extension ideas

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Ranked roughly by value for effort. Nothing here is committed to; the top three are the ones I would actually suggest.

| IDEA                        | EFFORT | VALUE | NOTES                                                                                                                                                                                                                                    |
|-----------------------------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calendar subscription (ICS) | Low    | High  | Serve a webcal feed so the sales appear directly in Google or Apple Calendar alongside everything else. This is the most natural possible feature for a calendar app, needs no new storage, and reaches the phone without push or email. |

|                                 |        |        |                                                                                                                                                                                                                                                         |
|---------------------------------|--------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscribe to one sale</b>    | Low    | High   | Reminders are currently country plus category. Letting someone follow just Black Friday, rather than everything tagged electronics, matches how people actually think. The detail drawer already preselects categories, so the interface is half built. |
| <b>Subscriber data backup</b>   | Low    | High   | Addresses the risk above: a nightly copy of backend/data, plus an export command. Small job, removes the only unrecoverable failure in the system.                                                                                                      |
| <b>Live deals strip</b>         | Medium | Medium | Pull the Pepper-network RSS from mydealz and Preisjaeger into a "hot right now" section, answering what is discounted today rather than which season is running. Depends on Q2.                                                                         |
| <b>Best time to buy</b>         | Medium | Medium | A view that inverts the calendar: given a category, when is it historically cheapest? Skis in March, electronics in late November. Derivable from the existing dataset with no new data.                                                                |
| <b>Shareable event links</b>    | Low    | Medium | A URL per sale so a specific window can be sent to someone. Also makes the site indexable, which it currently is not, being entirely client-rendered.                                                                                                   |
| <b>Weekly digest option</b>     | Medium | Medium | Depends on Q4. Would want a preference on the subscription and a second scheduled job.                                                                                                                                                                  |
| <b>More countries</b>           | Medium | Medium | France and Belgium first, since both publish binding national dates. Depends on Q5.                                                                                                                                                                     |
| <b>Retailer-level events</b>    | High   | Medium | MediaMarkt, Zalando and similar run their own recurring campaigns. Considerably more data to maintain, and much of it unsourceable - it risks the accuracy standard the rest of the dataset holds to.                                                   |
| <b>Browser end-to-end tests</b> | Medium | Low    | The date engine is well covered by 51 unit tests; the interface is not covered at all. Worth it only once the feature set stops moving.                                                                                                                 |

## 4. Suggested next actions

In order, assuming no answers to the questions above yet.

- Confirm the deploy went green, then run: `docker compose exec backend node send-test.js` - this is the only outstanding check on whether the Gmail app password works.
- Generate the VAPID keys on the NAS and enable push, then install the app on your phone and confirm a real notification arrives. That closes the last unverified feature.
- Look at light mode on a real screen and tell me what is wrong with it.
- Decide Q1, because it has a deadline the others do not: the 2027 Italian and South Tyrolean dates are published in late December, and until they are entered the app will quietly show estimates.
- Add the subscriber data backup - it is a small job guarding the only unrecoverable failure.

**A note on what this project optimises for**

The dataset is the product and the interface is a viewer for it. Every event carries a precision flag separating legally binding dates from mere tradition from pure marketing, and every event cites its sources - enforced by a test, not by convention.

That standard is worth protecting when adding features. A retailer-level dataset or a scraped refresh would both add breadth while quietly lowering the accuracy that makes the thing worth using.